ClipSAT : Cambridge & Edexcel Additional Mathematics 9606/4PM1 [ANSWER KEY]

Unit 7.01: Differentiation & Rates of Change

Q1. Differentiate with respect to x : y = 3x^{3} - 5x + 7 .

[ ] A: frac{dy}{dx} = 3x^{2} - 5
[ ] B: frac{dy}{dx} = 9x^{2}
[ ] C: frac{dy}{dx} = 9x^{3} - 5
[x] D: frac{dy}{dx} = 9x^{2} - 5

Solution: Apply the power rule frac{d}{dx}[x^n] = n x^{n-1} : frac{dy}{dx} = 33x^{2} - 5 = 9x^{2} - 5 .

Q2. Differentiate with respect to x : y = 6x^{4} - 5x + 7 .

[x] A: frac{dy}{dx} = 24x^{3} - 5
[ ] B: frac{dy}{dx} = 24x^{4} - 5
[ ] C: frac{dy}{dx} = 6x^{3} - 5
[ ] D: frac{dy}{dx} = 24x^{3}

Solution: Apply the power rule frac{d}{dx}[x^n] = n x^{n-1} : frac{dy}{dx} = 64x^{3} - 5 = 24x^{3} - 5 .

Q3. Differentiate with respect to x : y = 9x^{5} - 5x + 7 .

[x] A: frac{dy}{dx} = 45x^{4} - 5
[ ] B: frac{dy}{dx} = 45x^{5} - 5
[ ] C: frac{dy}{dx} = 9x^{4} - 5
[ ] D: frac{dy}{dx} = 45x^{4}

Solution: Apply the power rule frac{d}{dx}[x^n] = n x^{n-1} : frac{dy}{dx} = 95x^{4} - 5 = 45x^{4} - 5 .

Q4. Differentiate with respect to x : y = 12x^{6} - 5x + 7 .

[ ] A: frac{dy}{dx} = 72x^{5}
[ ] B: frac{dy}{dx} = 12x^{5} - 5
[ ] C: frac{dy}{dx} = 72x^{6} - 5
[x] D: frac{dy}{dx} = 72x^{5} - 5

Solution: Apply the power rule frac{d}{dx}[x^n] = n x^{n-1} : frac{dy}{dx} = 126x^{5} - 5 = 72x^{5} - 5 .

Q5. Differentiate with respect to x : y = 15x^{7} - 5x + 7 .

[ ] A: frac{dy}{dx} = 105x^{6}
[ ] B: frac{dy}{dx} = 15x^{6} - 5
[ ] C: frac{dy}{dx} = 105x^{7} - 5
[x] D: frac{dy}{dx} = 105x^{6} - 5

Solution: Apply the power rule frac{d}{dx}[x^n] = n x^{n-1} : frac{dy}{dx} = 157x^{6} - 5 = 105x^{6} - 5 .

Q6. Differentiate with respect to x : y = 18x^{8} - 5x + 7 .

[ ] A: frac{dy}{dx} = 144x^{7}
[x] B: frac{dy}{dx} = 144x^{7} - 5
[ ] C: frac{dy}{dx} = 144x^{8} - 5
[ ] D: frac{dy}{dx} = 18x^{7} - 5

Solution: Apply the power rule frac{d}{dx}[x^n] = n x^{n-1} : frac{dy}{dx} = 188x^{7} - 5 = 144x^{7} - 5 .

Unit 7.01: Differentiation & Rates of Change

Q7. Find the equation of the tangent line to the curve y = 2x + 1^3 at the point where x = 0 .

[ ] A: y = 2x + 1

[ ] B: y = 6x - 1

[x] C: y = 6x + 1

[ ] D: y = 6x + 3

Solution: At x = 0 , y = 1^3 = 1 . Derivative: frac{dy}{dx} = 32x+1^2 cdot 2 = 62x+1^2 . At x = 0 , slope m = 6 . Equation: y - 1 = 6x - 0 implies y = 6x + 1 .

Q8. Find the equation of the tangent line to the curve y = 3x + 1^3 at the point where x = 0 .

[ ] A: y = 9x + 3

[ ] B: y = 9x - 1

[ ] C: y = 3x + 1

[x] D: y = 9x + 1

Solution: At x = 0 , y = 1^3 = 1 . Derivative: frac{dy}{dx} = 33x+1^2 cdot 3 = 93x+1^2 . At x = 0 , slope m = 9 . Equation: y - 1 = 9x - 0 implies y = 9x + 1 .